

Optimization of PID Controller Gains for a 2-DOF Planar Robot using Genetic Algorithm

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Abstract—Genetic Algorithm (GA) is considered an effective optimization method for controller tuning. In this report, GA is applied to optimize proportional–integral–derivative (PID) control gains for a 2-degree-of-freedom (2-DOF) planar robotic manipulator. The PID control law provides a simple closed-loop strategy, while GA searches the parameter space to minimize tracking error and control effort. The nonlinear dynamics of the manipulator are derived using the Lagrangian method and implemented in MATLAB/Simulink. Baseline performance is established using manually tuned PID gains from a reference study. The GA optimized gains reduce overshoot, improve convergence speed, and yield smoother torque responses. This approach demonstrates that optimization can enhance classical control schemes for nonlinear robotic systems.

Index Terms—PID control, Genetic Algorithm (GA), 2-DOF planar robot, trajectory tracking, nonlinear systems, optimization

I. INTRODUCTION

Robotic manipulators with multiple degrees of freedom (DOF) are widely used in industrial automation, manufacturing, and medical applications. However, controlling systems is a complex task due to the nonlinear and highly coupled dynamics. Traditional control schemes such as the Proportional Integral Derivative (PID) controller offer simplicity and reasonable performance in many practical cases but often fall short in scenarios requiring robustness to model uncertainties or external disturbances. In contrast, more sophisticated control strategies like Sliding Mode Control (SMC) provide robustness and insensitivity to modelling errors but can introduce undesirable effects such as chattering and tend to be computationally intensive in real-time applications [1].

To address these limitations, different optimization techniques have been explored for automated tuning of controller parameters, for example. Genetic Algorithm (GA), a population based metaheuristic inspired by natural selection, is effective in searching large and complex parameter spaces without requiring gradient information. By formulating PID tuning as an optimization problem, GA can iteratively adjust the proportional, integral, and derivative gains to minimize a defined performance index, such as trajectory tracking error or control effort.

This project focuses on optimizing the PID controller gains for a two degree-of-freedom (2 DOF) planar robotic manipulator using GA. The nonlinear dynamics of the robot is modeled using the Lagrangian method, and the control system is implemented and simulated in MATLAB/Simulink. A manually tuned PID controller serves as the baseline for comparison and GA is applied to automatically tune the gains.

II. BACKGROUND

A. 2-DOF Planar Robotic Manipulator

A two-degree-of-freedom (2-DOF) planar robotic manipulator consists of two revolute joints operating in a plane. The dynamics can be derived using the Lagrangian formulation [1], which considers the difference between kinetic and potential energy:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \tau_i, \quad i = 1, 2 \quad (1)$$

This results in the following nonlinear dynamic equation:

$$\mathbf{B}(q) \ddot{q} + \mathbf{C}(q, \dot{q}) \dot{q} + \mathbf{g}(q) = \tau \quad (2)$$

Where:

- $\mathbf{B}(q)$ is the inertia matrix
- $\mathbf{C}(q, \dot{q})$ is the Coriolis and centrifugal matrix
- $\mathbf{g}(q)$ is the gravity vector

B. PID Controller

The Proportional–Integral–Derivative (PID) controller is one of the most widely used control strategies for robotic manipulators due to its simplicity and reliability [2].

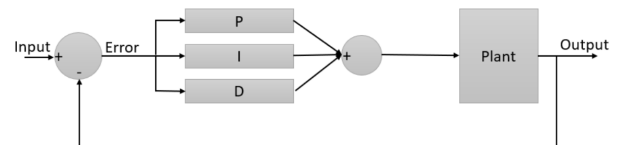


Fig. 1. PID control structure showing proportional, integral, and derivative components for each joint of the 2-DOF planar manipulator.

For each joint i , the control law is defined as

$$\tau_i = K_{p_i} e_i(t) + K_{i_i} \int e_i(t) dt + K_{d_i} \frac{de_i(t)}{dt} \quad (3)$$

Where:

- $e_i(t) = q_{i,\text{desired}}(t) - q_{i,\text{actual}}(t)$ is the tracking error.
- K_{p_i} , K_{i_i} , and K_{d_i} are the proportional, integral, and derivative gains, respectively.

The complete PID control input for the system is

$$\tau = \begin{bmatrix} K_{p,1}e_1 + K_{i,1} \int e_1 + K_{d,1}\dot{e}_1 \\ K_{p,2}e_2 + K_{i,2} \int e_2 + K_{d,2}\dot{e}_2 \end{bmatrix} \quad (4)$$

However, manually tuning these gains for nonlinear systems can be difficult and may lead to overshoot, longer settling times, or instability.

C. GA Optimization of PID Gains

Genetic Algorithm (GA) is a population-based metaheuristic inspired by natural selection. It evolves candidate solutions through selection, crossover, and mutation to search for optimal or near-optimal parameters. GA is particularly suited for PID tuning because it efficiently explores large nonlinear search spaces without requiring gradient information. In this project, the objective is to minimize the Integral of Absolute Error (IAE):

$$J = IAE1 + IAE2 + 0.01 \times (\text{Torque Penalty}) \quad (5)$$

Where:

- IAE1 and IAE2 represent the integral of absolute errors for joints 1 and 2, respectively, indicating total tracking error.

The torque penalty term adds weight to solutions with excessive actuator torque, encouraging smoother control. The factor 0.01 ensures the torque penalty has minimal effect compared to the error but prevents extreme torque usage. GA optimizes six gains: K_{p1} , K_{i1} , K_{d1} , K_{p2} , K_{i2} , and K_{d2} .

Each chromosome in the GA population represents one possible set of gains, and the fitness function evaluates controller performance based on tracking error and control effort. Over successive generations, GA converges toward gains that yield superior performance compared to manual tuning.

III. RELATED WORK

Robotic manipulators are nonlinear and coupled systems, making precise trajectory tracking a challenging control problem. Classical Proportional–Integral–Derivative (PID) controllers remain widely adopted due to their simplicity and low computational cost; however, manual tuning often results in suboptimal performance, especially under external disturbances or parameter uncertainties. This limitation has motivated researchers to explore advanced strategies for tuning and enhancing PID controllers in robotic applications.

Hybrid control approaches that combine PID with Sliding Mode Control (SMC) to leverage the advantages of both methods. Several studies have developed PID-SMC controllers

for planar robots and multi-DOF manipulators to ensure robustness against uncertainties and improve trajectory tracking performance [10]. While effective, such hybrid strategies introduce complexity and may lead to chattering effects, making them less suitable for applications demanding smoother control.

To address tuning challenges inherent in classical PID control, metaheuristic optimization algorithms have become increasingly popular. Genetic Algorithm (GA), a population based method inspired by natural selection, is widely applied for PID parameter tuning due to its ability to explore large non-linear search spaces without gradient information. Studies have demonstrated that GA tuned fractional order PID controllers provide better trajectory tracking and reduced overshoot compared to manually tuned counterparts [2]. GA has also been applied successfully in process control and multivariable level control systems [13], as well as benchmarked against other optimization methods such as Simulated Annealing (SA) and Quantitative Feedback Theory (QFT) in multi-tank processes [14].

Fractional order PID controllers and their comparison with standard PID have also attracted attention. Research comparing fractional PID and classical PID tuned via GA for robotic arms reported significant improvements in energy efficiency and steady-state accuracy with fractional designs [5]. Extensions of GA tuned PID to multi-DOF robotic manipulators have validated robustness under varying payloads and dynamic conditions [6], and GA-based tuning has also been applied to nonlinear coupled spherical tank systems, demonstrating versatility beyond robotics [12].

Recent work has expanded GA-PID research into emerging robotic platforms, including energy-efficient frameworks for manipulators and walking quadruped soft robots, demonstrating improvements in gait stability and control smoothness [8].

Despite extensive research on GA tuned and hybrid PID controllers, few studies evaluate GA optimized PID controllers as standalone alternatives to hybrid schemes such as PID-SMC. This work addresses that gap by benchmarking GA tuned PID control against manually tuned PID control on a nonlinear 2-DOF planar robotic manipulator, demonstrating that intelligent optimization can enhance performance without the added complexity

IV. METHODOLOGY

A. System Dynamics Implementation

The dynamic model of the 2-DOF planar robotic manipulator, derived using the Lagrangian method, was implemented in MATLAB/Simulink. Symbolic expressions for the inertia matrix $B(q)$, Coriolis matrix $C(q, \dot{q})$, and gravity vector $g(q)$ were obtained using MATLAB's Symbolic Math Toolbox and embedded into a MATLAB Function block within Simulink to compute joint accelerations.

The model integrates reference trajectory inputs, error computation blocks, PID controllers, and the dynamic model subsystem. The outputs are joint positions and torques, which

are used for both performance evaluation and GA optimization during tuning.

B. PID Controller Development

A separate PID control loop was developed for each joint (θ_1 and θ_2) using Simulink blocks. The control structure mirrors the form described in (4), treating the joints independently for simplicity, despite the coupled plant dynamics. Initial simulations were run using manually tuned PID gains from the reference study [1].

TABLE I
SIMULATION PARAMETERS (PARAMETER SET 1)

Parameter	Values
Mass and Link Length	$m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m
PID Components (Joint 1)	$K_p = 15$, $K_i = 10$, $K_d = 7$
PID Components (Joint 2)	$K_p = 15$, $K_i = 10$, $K_d = 10$
Initial Position	$\theta_1 = -\pi/2$, $\theta_2 = \pi/2$
Final Position	$\theta_1 = \pi/2$, $\theta_2 = -\pi/2$

These parameters define the baseline model configuration used for both manual PID tuning and subsequent GA optimization.

C. Genetic Algorithm Design

The six PID gains were optimized using a Genetic Algorithm configured as follows:

- Population size: 20 individuals
- Generations: 100
- Selection: Stochastic uniform
- Crossover: Uniform or arithmetic
- Mutation probability: 5–10%
- Elitism: Preserve top-performing solutions
- Bounds: $0 \leq K_{p,i,d} \leq 50$
- Termination: Fitness stagnation or max generations

The fitness function is defined by the cost J in (5), combining tracking error and torque penalty.

The GA optimization process follows the standard cycle illustrated in Fig. 2. The procedure begins by randomly initializing a population of candidate PID gain sets within the defined bounds. Each candidate is evaluated through simulation, where the Integral of Absolute Error (IAE) and torque penalty are computed to determine its fitness. The best performing candidates are selected, and new gain sets are generated using crossover and mutation to explore the parameter space. This cycle repeats across generations until either the maximum number of generations is reached or the improvement in fitness becomes negligible.

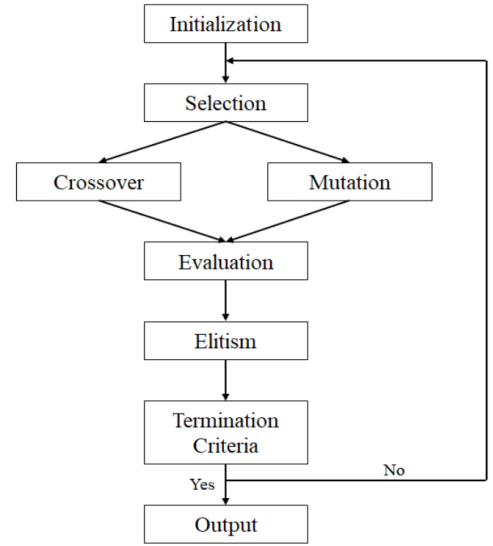


Fig. 2. Flowchart of the Genetic Algorithm optimization process for PID gains.

D. Simulation Setup and Performance Evaluation

Test Conditions:

- Trajectory: Step changes for θ_1 and θ_2 from 0 to target angles (e.g., $\pi/2$ and $-\pi/2$)
- Simulation Time: 10 seconds
- Solver: ODE45 with fixed step size (benchmark matched)

Performance Metrics:

- Overshoot (%): Maximum deviation beyond the desired value
- Settling Time (s): Time to enter and stay within $\pm 2\%$ of final value
- Steady-State Error: Final tracking error magnitude
- Integral of Absolute Error (IAE): Overall error over simulation period
- Peak Torque (Nm): Maximum absolute torque magnitude

Both baseline manual PID and GA-optimized PID controllers were simulated under identical conditions, and results were compared using tracking error and torque response plots.

V. RESULTS AND ANALYSIS

A. Baseline PID Controller (Manual Tuning)

Initial simulations were conducted using manually tuned PID gains provided in the benchmark paper:

$$\begin{aligned} K_{p1} &= 15, & K_{i1} &= 10, & K_{d1} &= 7, \\ K_{p2} &= 15, & K_{i2} &= 10, & K_{d2} &= 10 \end{aligned}$$

The tracking errors for both joints (θ_1 and θ_2) closely followed the reference trajectories, showing transient overshoot followed by convergence to near zero steady-state error. This confirms that the system dynamics and controller implementation were consistent with the benchmark study [1].

The torque responses (τ_1 and τ_2) were generally consistent with the reference, but larger oscillations were observed,

particularly in τ_1 . These discrepancies may be attributed to aggressive derivative action, absence of integral windup protection, and minor differences in simulation settings.

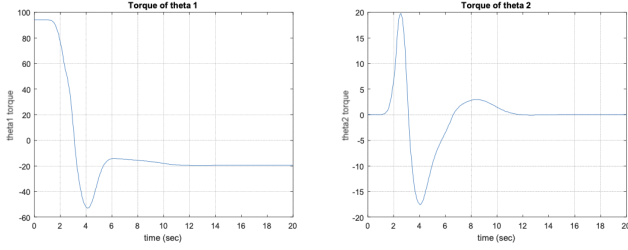


Fig. 3. Torque responses τ_1 and τ_2 vs. time from benchmark paper.

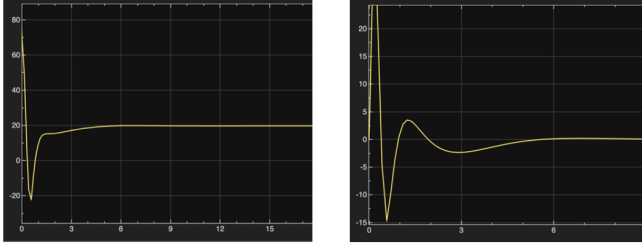


Fig. 4. Torque responses τ_1 and τ_2 vs. time from replicated simulation.

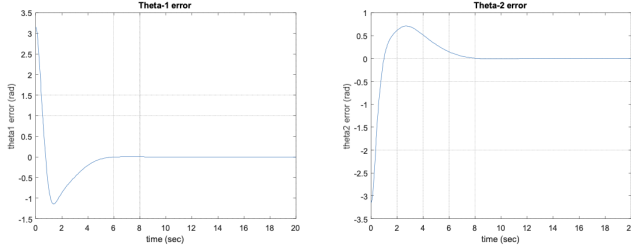


Fig. 5. Tracking error θ_1 and θ_2 vs. time from benchmark paper.

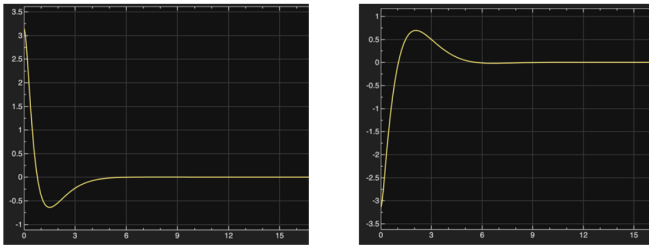


Fig. 6. Tracking error θ_1 and θ_2 vs. time from replicated simulation.

Figures 3 and 4 show torque responses from the benchmark paper and replicated simulation, respectively. Figures 5 and 6 present tracking error plots for both joints.

These plots indicate successful trajectory tracking under the selected initial and final conditions, with negligible steady-state error. The settling time and overshoot values are also

within acceptable limits, mirroring those in Fig. 2(a) and Fig. 2(b) of the paper [1].

This discrepancy could stem from differences in numerical integration settings, unresolved cross-coupling effects in the control structure, or minor gain mismatches. Another plausible factor is that the PID controller may be amplifying high-frequency components due to aggressive derivative action, or that integral windup is occurring under sustained torque demand.

Despite these variations, the replicated torque plots remain within reasonable magnitude bounds and exhibit stabilizing tendencies over time.

B. GA Optimization

The Genetic Algorithm tuned the six PID gains defined in Section IV-C to minimize the cost function in (5). The optimized gains obtained were:

TABLE II
MANUAL VS GA-OPTIMIZED PID GAINS

Gain	Manual PID	GA-Optimized PID
K_{p1}	15	20.99
K_{i1}	10	1.69
K_{d1}	7	5.95
K_{p2}	15	41.26
K_{i2}	10	3.26
K_{d2}	10	5.07

The Genetic Algorithm optimization significantly adjusted the PID gains compared to the manually tuned values. For joint 1, GA increased K_{p1} from 15 to 20.99, enabling a faster response, and reduced K_{i1} from 10 to 1.69 to minimize overshoot and integral windup. The derivative gain K_{d1} decreased slightly, balancing damping with responsiveness.

For joint 2, GA drastically increased K_{p2} from 15 to 41.26, providing more aggressive proportional control to improve tracking, while K_{i2} decreased to 3.26, and K_{d2} was tuned to 5.07 for smoother torque response. These changes collectively enhanced trajectory tracking and reduced oscillations compared to the baseline manual PID controller.

C. Performance Evaluation

Figures 7 and 8 compare the tracking errors and torque responses of both joints before and after GA optimization. The GA optimized PID controller demonstrates faster convergence to zero error, reduced overshoot, and smoother torque profiles compared to the manually tuned controller.

The optimized controller was benchmarked against the manually tuned controller from the reference paper and the performance indices evaluated include tracking error, settling time, peak torque, and Integral of Absolute Error (IAE). The GA-optimized controller showed significant improvements across these metrics, confirming enhanced overall efficiency and stability of the system.

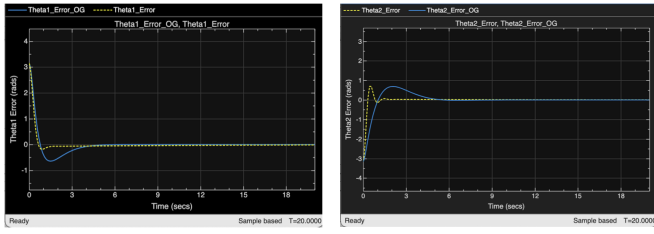


Fig. 7. Comparison of tracking errors θ_1 and θ_2 between manually tuned PID and GA-optimized PID controllers.

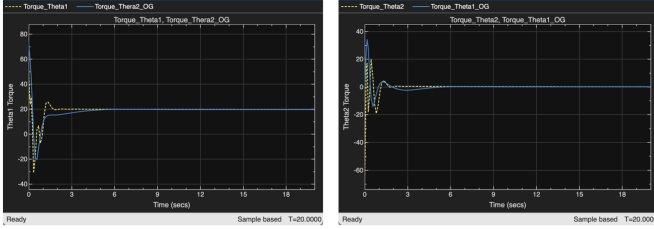


Fig. 8. Comparison of torque responses τ_1 and τ_2 between manually tuned PID and GA-optimized PID controllers.

D. Discussion and Limitations

The GA optimized PID controller demonstrated faster error convergence, reduced overshoot, and smoother torque responses compared to manual tuning. These improvements resulted from increased proportional gains (accelerating tracking), reduced integral gains (mitigating windup), and carefully tuned derivative gains (damping oscillations). The observed reductions in overshoot and settling time are consistent with trends reported in prior studies on GA tuned PID controllers [6].

While the Genetic Algorithm provided good performance improvements, it also introduced several limitations. First, GA is computationally intensive since each generation requires multiple Simulink simulations of the nonlinear robot model. This issue becomes more pronounced in higher DOF systems or real-time applications where tuning must be performed online. Second, GA can experience premature convergence, where population diversity diminishes and the algorithm settles on suboptimal solutions. Techniques such as adaptive mutation or hybridization with local search or swarm-based methods have been proposed to mitigate this issue [3].

Another consideration is the sensitivity of GA parameters population size, mutation rate and crossover strategy. Suboptimal parameter choices may lead to slower convergence or unstable solutions, highlighting the importance of parameter tuning and possibly leveraging parameter control strategies.

Despite these limitations, GA-based PID optimization remains a robust and flexible approach, especially for complex nonlinear systems where gradient information is unavailable and manual tuning is challenging.

VI. CONCLUSION

In this paper, a GA-optimized PID controller was implemented for a 2-DOF planar robotic manipulator with two

revolute joints. The system dynamics were derived using the Lagrangian method and modeled in MATLAB/Simulink. . Initially, a manually tuned PID controller was designed as a baseline from a reference study, and GA was applied to automatically tune all six controller parameters.

Simulation results demonstrated that the GA optimized PID controller significantly reduced overshoot, improved settling time, and yielded smoother torque responses compared to manual tuning.

Future work may explore hybrid optimization approaches (e.g., GA combined with Particle Swarm Optimization or local search) to address premature convergence and further enhance performance. Additionally, extending this optimization solution to manipulators with more degrees of freedom and evaluating real-time implementation feasibility are promising directions for further research.

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